CSE3433
SOFTWARE ARCHITECTURE
PROJECT PROPOSAL

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PROJECT TITLE
Event-Driven Job Requirement & Application Processing

ARCHITECTURE STYLE
In this rapidly developing digital era, traditional monolithic systems are no longer able to accommodate the needs of modern applications that require fast response and high flexibility. Therefore, this project chose Event-Driven Architecture (EDA) as the basis for system design. EDA allows the system to respond automatically to each event such as “Application Submitted” or “Interview Scheduled”. This is because the communication is loosely coupled, the system becomes more flexible, scalable, and able to respond in real time. This approach is very suitable for recruitment systems that involve many processes that occur continuously

PROBLEM DESCRIPTION
The current recruitment process still faces various constraints such as delays in processing applications, lack of integration between systems, and lack of real-time notifications to users. This situation causes candidates to often not receive clear feedback. This causes HR to face difficulties in managing application data in an orderly and systematic manner. In addition, most of the traditional systems used are inflexible and difficult to expand. For example, when the number of users increases. This can cause system performance to deteriorate and increase the risk of errors in data management. Therefore, there is a need for a system that is more responsive, integrated, and capable of supporting the recruitment process in a more efficient and modern way.

SYSTEM OBJECTIVES
1. To provide an Event-Based Recruitment System. The system will allow to act automatically when an event occurs. 2. To improve the Efficiency of the Recruitment Process. The system will reduces dependence on manual processes. 3. To provide Real-Time Updates. The system will provide instant notifications to users (HR and candidates). 4. Create a Modular and Scalable System. Each component can be developed and improved separately. 5. Improve User Experience. The system will provide a more responsive and easy-to-use.

FUNCTIONAL REQUREMENT
1. User Management • Users can register as a job seeker or employer. • Login and logout of the system. • User profile management. 2. Job Posting Module • HR can create, update and delete job posting. • Event: JobPosted, JobUpdated 3. Job Application Module • Job seekers can apply for a job and also can upload their resume. • Event: ApplicationSubmitted 4. Application Processing Module • The system will save the application and evaluate the application on a basic basis. For example keyword matching. • Event: ApplicationReviewed 5. Notification Module • The system sends notifications when application is accepted, status changes and also when interview is scheduled. • Event: NotificationSent 6. Interview Scheduling Module • HR can set an interview date. • Event: InterviewScheduled 7. Dashboard Module • HR can view candidate list and application status. • Job seeke can view their application status.

TEAM MEMBER ROLES
1. Adam: Project Manager & System Architect Responsibilities: • Manage project planning and timeline • Determine system requirements (requirements analysis) • Design system architecture using Event-Driven Architecture • Planning event flow between modules • Provide system documentation 2. Radilan: Full Stack Developer & Tester Responsibilities: • Developing the frontend (user interface) • Develop backend (system logic & event handling) • Integrate all system modules • Conduct system testing (testing) • Fix bugs and keep the system working properly